

Apostol Chapter 1 - Exercise 5

Afonso Vilhena Francisco

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If $(a, b) = 1$, then $(a + b, a^2 - ab + b^2)$ is either 1 or 3.

Suppose that $(a, b) = 1$.

Let $d := (a + b, a^2 - ab + b^2)$. Then, by definition, $d | (a + b)$ and $d | (a^2 - ab + b^2)$. In particular, d divides any integer linear combination of the previous ones, that is,

$$d | (a + b)x + (a^2 - ab + b^2)y, \quad \text{for any integers } x \text{ and } y$$

Taking $x = a - 2b$ and $y = -1$, we have that d divides

$$\begin{aligned} (a + b)(a - 2b) - (a^2 - ab + b^2) &= a^2 - 2ab + ab - 2b^2 - a^2 + ab - b^2 \\ &= -3b^2 \end{aligned}$$

Therefore, d also divides $3b^2$, that is, $3b^2 = dq_1$ for some integer q_1 .

In a similar way, taking $x = b - 2a$ and $y = -1$, we have that d divides

$$\begin{aligned} (a + b)(b - 2a) - (a^2 - ab + b^2) &= ab - 2a^2 + b^2 - 2ab - a^2 + ab - b^2 \\ &= -3a^2 \end{aligned}$$

Therefore, d also divides $3a^2$, that is, $3a^2 = dq_2$ for some integer q_2 .

Since $(a, b) = 1$, then, by *Exercise 3*, we have that $(a^2, b^2) = 1$. That is,

$$1 = a^2u + b^2v,$$

for some integers u and v . Multiplying the previous equation by 3 and substituting the terms $3a^2$ and $3b^2$, we get

$$\begin{aligned} 3 &= (3a^2)u + (3b^2)v \\ &= dq_2u + dq_1v \\ &= d(q_2u + q_1v) \end{aligned}$$

Therefore, $d | 3$, so it must be the case that $d \in \{1, 3\}$.